Brain Tumour Magnetic Resonance Image Classification with Perceptual-Hash Deduplication: A Comparative Study Using Advanced Deep Learning Models

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ABSTRACT

Magnetic resonance imaging (MRI) is the standard investigation for brain tumours, but published deep learning accuracies are difficult to compare because studies differ in data, splitting and evaluation, and near-duplicate images from shared public sources leak between training and test partitions. We introduce a leakage-controlled, calibrated, and efficient pipeline for MRI tumour detection and subtype classification that discards near-duplicates and conflicting images prior to group-aware splitting. We evaluate nine ImageNet-pretrained convolutional, transformer and hybrid backbones under an equal budget of optimiser steps using bootstrapped confidence intervals, McNemar tests with Holm correction, temperature scaling, and Grad-CAM and background-only shortcut audits. The engineering application is the selection of a deployable tumour detector and subtype classifier. Deduplication reduced the binary detection pool from 3,253 to 1,185 images and the subtype classification pool from 5,828 to 4,730, and curation ran at about 85 images per second on four CPU cores. On 171 binary detection test images, CoAtNet-0, EfficientNetB0 and ViT-B/16 reached 99.42% accuracy (95% confidence interval 98.25–100%), and no pair of models differed significantly. The subtype classification grouping left only 45 test images, on which six models were error-free, so they cannot rank the models. Models retrained with the head masked out still exceeded the majority-class rate for 7 of 9 binary detection and all 9 subtype classification backbones, reaching 83.04% and 97.78% accuracy, so these public collections carry label information outside the brain. EfficientNetB0 was selected for deployment because it shares the top binary detection accuracy with only 4.01 million parameters, less than a sixth of CoAtNet-0 and a twentieth of ViT-B/16.

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